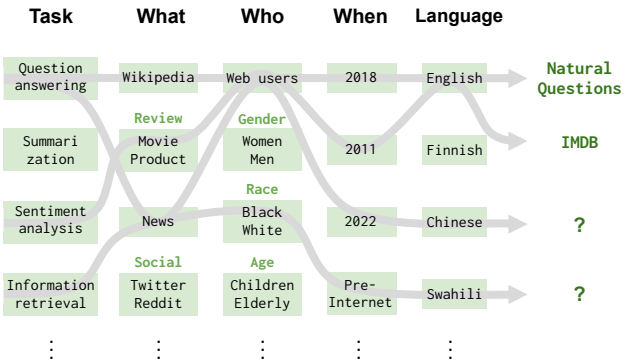

Figure 8: **Scenario structure.** Scenarios are what we want the language model to do. To specify a scenario, we break it down into a *task*, *domain*, and *language*, further subdividing the domain into properties of the text (*what*), speaker (*who*), and the time/circumstances (*when*). Examples of scenarios include (question answering, (clinical notes, doctors, now), English) and (toxicity detection, (tweets, Egypt, Internet-era), Arabic).

characterizes the type of data we want the system to do well on; and (iii) the *language* or language variety (e.g. Spanish). Tasks, domains, and languages are not atomic or unambiguous constructs: they can be made coarser and finer, but we use them as intuitive *structure* for the space of scenarios. Given this structure, we deliberately select scenarios based on three overarching principles: (i) coverage of the space, (ii) minimality of the set of selected scenarios, and (iii) prioritizing scenarios that correspond to user-facing tasks. Alongside feasibility given our resources (which we explicitly acknowledge), this defines the core scenarios we evaluate on, which we will measure all metrics on. In §10.1: MISSING-SCENARIOS, we highlight regions of the scenario space that we taxonomize but do not currently cover in our benchmark/scenario selection.

3.1 Taxonomy

Track	Tasks
Computational Social Science and Cultural Analytics	No canonical tasks/not task-centric
Dialogue and Interactive Systems	Chit-chat dialogue, task-oriented dialogue
Discourse and Pragmatics	Discourse parsing, sentence ordering, coreference resolution
Ethics and NLP	Toxicity and hate speech detection, misinformation and fake news detection
Generation	Data-to-text generation,
Information Extraction	Named entity recognition, entity linking, entity extraction, relation extraction, event extraction, open information extraction
Information Retrieval and Text Mining	Information retrieval and passage retrieval
Interpretability and Analysis of Models for NLP	No canonical tasks/not task-centric
Language Grounding to Vision, Robotics and Beyond	Image captioning, visual question answering, instruction following, navigation
Linguistic Theories, Cognitive Modeling, and Psycholinguistics	No canonical tasks/not task-centric
Machine Learning for NLP	Language modeling
Machine Translation and Multilinguality	Machine translation
NLP Applications	No canonical tasks
Phonology, Morphology, and Word Segmentation	Tokenization, lemmatization,
Question Answering	Question answering and reading comprehension
Resources and Evaluation	No canonical tasks/not task-centric
Semantics: Lexical	Word sense disambiguation, word sense induction
Semantics: Sentence-level Semantics, Textual Inference, and Other Areas	Semantic parsing, natural language inference, semantic role labeling/slot filling, semantic textual similarity, paraphrase detection
Sentiment Analysis, Stylistic Analysis, and Argument Mining	Sentiment analysis, style transfer, argument mining, stance detection, opinion mining, text simplification
Speech and Multimodality	Text-to-speech, speech-to-text
Summarization	Summarization, sentence compression
Syntax: Tagging, Chunking and Parsing	POS tagging, chunking, constituency parsing, dependency parsing, grammar induction, grammatical error correction

Table 1: **Taxonomy of tasks.** To taxonomize the space of tasks, we leverage the NLP community's taxonomy of subareas as codified by the ACL 2022 list of tracks. For each track, we then expand it into canonical tasks associated with that track.

Tasks. Given the ubiquity of natural language, the field of natural language processing (NLP) considers myriad tasks that correspond to language's many functions (Jurafsky & Martin, 2000). It is difficult to derive a space of tasks from first principles, so we compile existing sources of tasks. Naturally, given NLP is a task-centric field, we begin with tasks that have been extensively studied by the NLP community. To generate this set, we take the tracks at a major NLP conference (ACL 2022), which reflect the "relevant

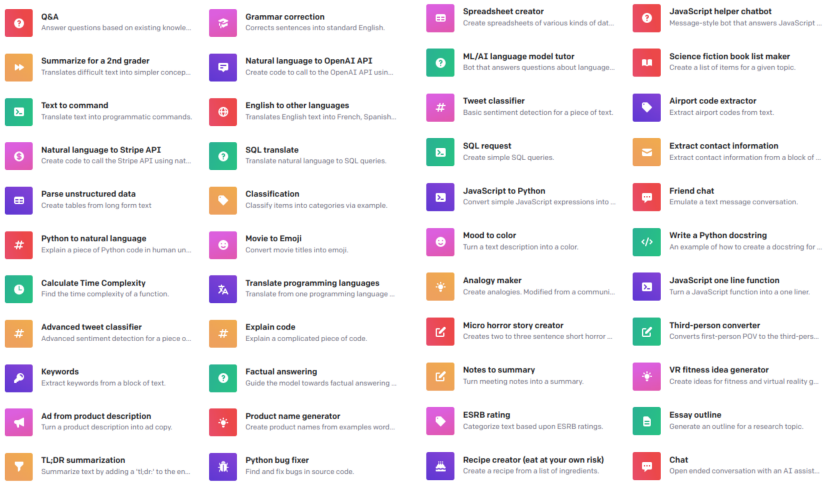

Figure 9: **Modern use cases for language models.** An assortment of (largely novel/historically unexplored) potential use cases for language models. Figure sourced from `https://beta.openai.com/examples/`.

topics" of study in NLP at the time of writing.[32] For each track, we map the associated subarea of NLP to canonical tasks for that track in Table 1. We acknowledge there is some subjectivity in choosing what is "canonical", which was only done so as to make this process manageable.

While these tasks often have long traditions of study in the NLP research community, we make two observations: (i) these tasks often have important intra-task structure (e.g. we refer to all of question answering as one "task", whereas the QA community likely would further decompose QA into finer-grained categories (Rogers et al., 2021)) and (ii) while these tasks have (long) traditions of study in NLP research, they are not the only, or even the most societally/economically impactful, tasks.

For example, the deployment of language models as interfaces by OpenAI, Cohere, and AI21 Labs has introduced use cases beyond what the NLP community has historically studied (see Figure 9 and compare to Table 1). In fact, some of these tasks are fundamentally new: the advent of sufficiently capable technology motivates the consideration of tasks that were not previously conceived (or conceived as within scope for algorithmic systems). Further, these tasks pattern quite differently from what has been traditionally studied in the NLP and AI research communities (see Ouyang et al., 2022). This introduces a fundamental challenge to stating the space of tasks: we will not be able to conceive of the true full space of tasks until we see technology that makes us consider these tasks. And, more broadly, even articulating (let alone covering) the long tail of known potential use cases remains open.

Domains. Domains are a familiar construct in NLP, yet their imprecision complicates systematic coverage of domains. We further decompose domains according to 3 W's:

1. **What** (genre): the type of text, which captures subject and register differences. Examples: Wikipedia, social media, news, scientific papers, fiction.

2. **When** (time period): when the text was created. Examples: 1980s, pre-Internet, present day (e.g. does it cover very recent data?)

3. **Who** (demographic group): who generated the data or who the data is about. Examples: Black-/White, men/women, children/elderly.

[32] `www.2022.aclweb.org/callpapers`

World Languages
Kalaallisut
Svenska
Vepsän kel'
Íslenska
Yupik
Scots
Deutsch
ненэця' вада
Русский
ᐃᓄᒃᑎᑐᑦ
Français
Italiano
Türkçe
ᓀᐦᐃᔑᓂᐧᐃᐣ
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